SubGround: Retrieving Query-Specific Subgraphs to Ground Language Model Answers

Ravi Menon

Northbridge University

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Abstract

SubGround retrieves a query-specific subgraph before a Large Language Model writes an answer. A Graph Neural Network scores nodes and edges in a Knowledge Graph, selecting a compact set of explicit facts rather than a long undifferentiated context. The subgraph is serialized into the prompt and used for Multi-hop QA on HotpotQA and Hallucination Reduction evaluation on FactProbe. SubGround provides a modest EM improvement on HotpotQA and cuts the FactProbe hallucination rate roughly in half compared with plain retrieval-augmented generation. The results suggest that structured evidence can improve both reasoning and factual discipline, although graph incompleteness still limits recall.

Keywords

Subgraph grounding, Multi-hop QA, Hallucination Reduction, FactProbe

1. Introduction

Passage retrieval supplies evidence but does not guarantee that the evidence is connected or sufficient. For Multi-hop QA, a system may need to traverse several facts; for Hallucination Reduction, it must also distinguish supported from unsupported claims. SubGround uses a Knowledge Graph to make this evidence explicit. It builds on RelForge in [1], contrasts with passage retrieval in [2], and reuses the FactProbe evaluation in [3].

2. Method

A question encoder maps the query to an initial node distribution over the Knowledge Graph. A Graph Neural Network propagates this signal across typed edges for three message-passing steps. Node and edge scores are combined with source confidence and a hop penalty, then a beam search extracts a small connected subgraph. Each selected fact retains its source passage and identifier. The subgraph serializer emits one normalized fact per line followed by a short instruction to use only supported facts. A Large Language Model receives this serialization and produces the answer. For HotpotQA, training rewards answer-bearing paths; for FactProbe, a claim-supporting subset is used without changing the generator. The system reports both answer quality and unsupported-span rate, separating retrieval failures from generation failures.

3. Experiments

On HotpotQA, plain retrieval-augmented generation scores 31.2 EM, while SubGround reaches 34.8 EM. On FactProbe, the hallucination rate falls from 18.6% to 9.4%. The gain is largest when the answer requires two or three linked facts. Failures mainly arise when the needed relation is absent from the Knowledge Graph.

System	HotpotQA EM	FactProbe hallucination
Plain retrieval-augmented generation	31.2	18.6%
SubGround	34.8	9.4%

4. Conclusion

SubGround demonstrates that a Graph Neural Network can select compact, query-specific evidence for a Large Language Model. It improves Multi-hop QA and materially advances Hallucination Reduction, while leaving graph coverage as the central limitation. Future work should combine structured and textual evidence without losing traceability.

References

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